

Explanation of Student Grades Recording System

This system is a simple C program designed to read student names and their grades, validate the input, and write the collected data into a CSV file.

Key Components:

1. Conditional Branching (if):

- The ``if`` statement is used to ensure that the input grade is within the valid range (between 0.0 and 100.0). If the input is invalid, the program prompts the user to re-enter a correct value.

2. Repetition (while):

- A ``while`` loop is used to allow the user to input multiple records of student names and grades. The loop continues until the user decides to stop by entering 'n' (for no) when prompted.

3. User-Defined Function (save_to_csv):

- The function ``save_to_csv`` is responsible for writing the data into the CSV file called "students.csv". It takes the student's name and grade as input parameters and appends them to the CSV file.

4. Input from Keyboard (scanf):

- The ``scanf`` function is used to read the user's input. It captures the student's name, grade, and whether they want to continue entering more data.

5. Output to CSV:

- The program writes student data into a CSV file, where each record consists of a student's name and their grade, formatted in "name,grade" format.

Program Flow:

- The user is prompted to input a student's name and grade.
- The system validates the grade and prompts the user again if it is outside the valid range.
- The data is then written into the "students.csv" file.
- The user is asked whether they want to enter another record. If they choose 'n', the program exits. If they choose 'y', it loops back to enter another student's data.

The CSV file can then be opened in any spreadsheet software like Excel or a text editor for review.

Example CSV Output:

...

John,88.50

Emily,92.00

Michael,76.25

...

This simple system is a useful practice for handling input validation, file handling, and loop structures in C programming.